rate of movement of people from second to first floor, and hence the number of people in each floor will remain the same. Thus the population of people on the two floors is in a dynamic equilibrium Let us extend this analogy to understand dynamic nature of equilibrium.

Chemical reactions which are reversible do not cease, when equilibrium is attained. At equilibrium the forward and the backward reactions are proceeding at the same rate and no macroscopic change is observed. So chemical equilibrium is in a state of dynamic equilibrium.

For example,

$$H_2\,(g) + I_2\,(g) \rightleftharpoons 2HI\,(g)$$

## 8.5 Homogeneous and heterogeneous equilibria

### 8.5.1 Homogeneous equilibrium

In a homogeneous equilibrium, all the reactants and products are in the same phase.

For example:

$$H_2\,(g) + I_2\,(g) \rightleftharpoons 2HI\,(g)$$

In the above equilibrium, $H_2$, $I_2$ and HI are in the gaseous state.

Similarly, for the following reaction, all the reactants and products are in homogeneous solution phase.

$$CH_3COOCH_3(aq) + H_2O(aq) \rightleftharpoons CH_3COOH(aq) + CH_3OH\,(aq)$$

### 8.5.2 Heterogeneous equilibrium

If the reactants and products of a reaction in equilibrium, are in different phases, then it is called as heterogeneous equilibrium.

Example:

$$H_2O\,(l) \rightleftharpoons H_2O\,(g)$$

$$CaCO_3\,(s) \rightleftharpoons CaO\,(s) + CO_2\,(g)$$

## 8.6 Law of mass action

In 1864 two Norwegian chemists namely Maximilian Guldberg and Peter Waage formulated the law of mass action, based on the experimental studies of many reversible reactions. The law states that,

"At any instant, the rate of a chemical reaction at a given temperature is directly proportional to the product of the active masses of the reactants at that instant".

$$\text{Rate } \alpha\ [\text{Reactant}]^x$$

where, x is the stoichiometric coefficient of the reactant and the square bracket represents the active mass (concentration) of the reactants.

$$\text{Active mass} = \left(\frac{n}{V}\right) \text{ mol dm}^{-3}\text{(or) mol L}^{-1}$$



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